Phishing Email Detection & Awareness Report

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Executive Summary

This report presents an analysis of phishing email attacks and provides awareness guidelines to help users identify and avoid such threats. Phishing attacks are one of the most common methods used by cybercriminals to steal sensitive information, including passwords, financial data, and personal details. This assessment analyses sample phishing emails, identifies key indicators of malicious intent, and explains how these attacks work simply and understandably. The report also provides practical recommendations and user awareness guidelines to reduce the risk of phishing attacks in a business environment.

Objective

The objective of this report is to:
- Analyse phishing email samples
- Identify common phishing indicators
- Classify email risks
- Educate users on how phishing attacks work
- Provide actionable prevention strategies

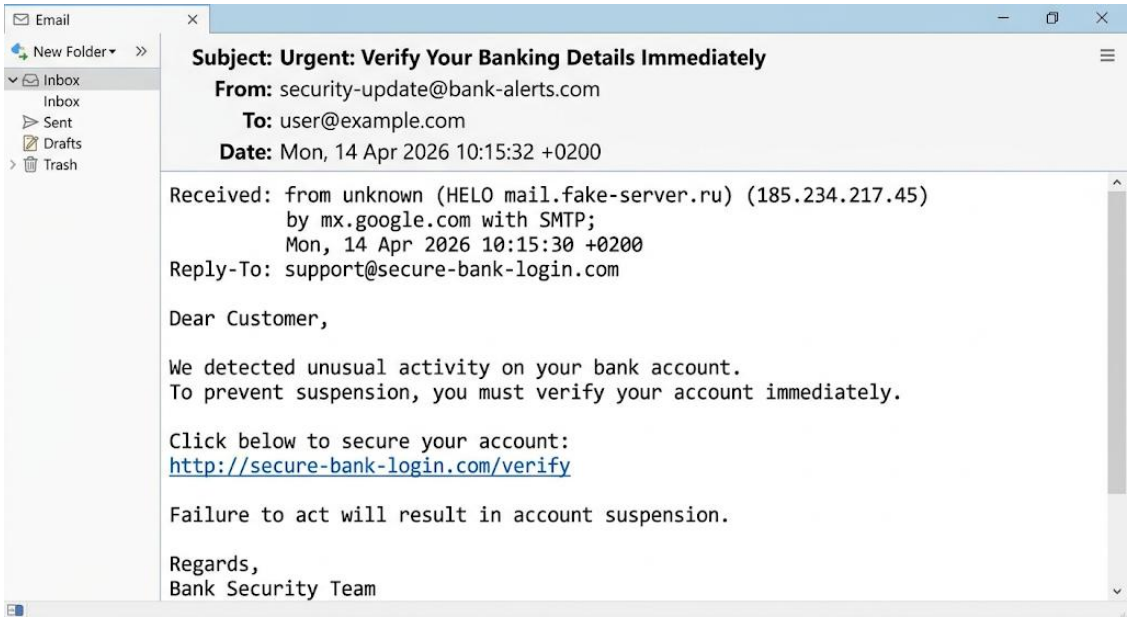
Methodology

The analysis was conducted using a structured approach:
- Collection of phishing email samples
- Inspection of email content and structure
- Analysis of suspicious links and domains
- Review of email headers using online tools
- Identification of phishing indicators
- Classification of risk levels

No interaction with malicious links or harmful actions was performed during this assessment.

Phishing Email Analysis

- **Sample Email**



▪ Identified phishing indicators

1. Suspicious Sender Domain
The email does not clearly identify a legitimate organization.
2. Urgency and Fear Tactics
The message pressures the user to act quickly ("within 24 hours").
3. Suspicious Link
The URL provided is not a trusted domain and may lead to a malicious website.
4. Generic Greeting
The email uses "Dear User" instead of the recipient's name.
5. Lack of Personalization
No account-specific details are provided.

▪ Risk classification

Risk Level: PHISHING (High Risk)
This email shows multiple strong indicators of phishing, including header manipulation, domain spoofing, and social engineering techniques.

▪ Analysing email headers

- The email was sent from an unknown server (mail.fake-server.ru), not a legitimate bank server.
- The IP address (185.234.217.45) is suspicious and not linked to a trusted organization.
- The sender domain (bank-alerts.com) is fake and not an official bank domain.
- The Reply-To address (secure-bank-login.com) is different from the sender, indicating possible redirection to a malicious site.

MessageId	1234567890@mail.fake-server.ru
Created at:	4/14/2026, 10:15:32 AM GMT+2 (Delivered after -2 sec)
From:	"Bank Security" <security-update@bank-alerts.com>
To:	user@example.com
Subject:	Urgent: Verify Your Banking Details Immediately
SPF:	fail with IP Unknown! Learn more
DKIM:	fail with domain Unknown! Learn more
DMARC:	fail Learn more

#	Delay	From *			To *	Protocol	Time received
0	-2 sec	mail.fake-server.ru	→	[Google]	mx.google.com	ESMTPS	4/14/2026, 10:15:30 AM GMT+2

How the attacks work

Phishing attacks work by tricking users into believing that an email is from a trusted source. In this case, the attacker creates a sense of urgency to force the user to click a malicious link. Once clicked, the user may be redirected to a fake login page designed to steal credentials. The attacker can then use the stolen information to gain unauthorized access to accounts, commit fraud, or launch further attacks.

Awareness & Prevention Guidelines

To protect against phishing attacks, users should:
- Always verify the sender's email address
- Avoid clicking links from unknown or suspicious emails
- Hover over links to check the actual URL
- Look for spelling and grammatical errors
- Be cautious of urgent or threatening language
- Never share passwords or sensitive information via email

Dos and Don'ts

Dos:
- Verify emails before acting.
- Report suspicious emails to IT/security teams.
- Use strong and unique passwords.
- Enable multi-factor authentication (MFA).

Don'ts:
- Do not click on unknown links.
- Do not download unexpected attachments.

- Do not respond to suspicious emails.
- Do not share personal or login information.

Conclusion

Phishing attacks remain a significant threat to individuals and organizations. This report demonstrated how phishing emails can be identified through common indicators such as suspicious links, urgency tactics, and lack of personalisation. By increasing awareness and following recommended security practices, users can significantly reduce the risk of falling victim to phishing attacks. Continuous education and vigilance are key to maintaining a secure environment.